

**KONFIGURASI MIKROTIK ROUTEROS SEBAGAI GATEWAY
INTERNET, DHCP SERVER, NAT, DAN FIREWALL PADA JARINGAN
VIRTUAL MENGGUNAKAN VIRTUALBOX**



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Kelas: TI4B

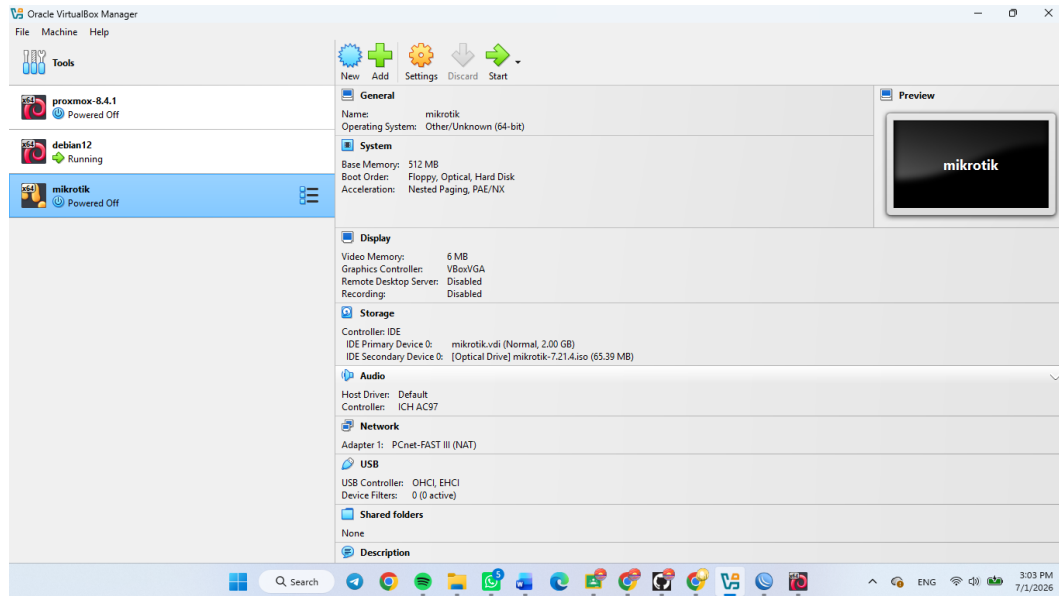
NIM: 2402043

PROGRAM STUDI D3 TEKNIK INFORMATIKA

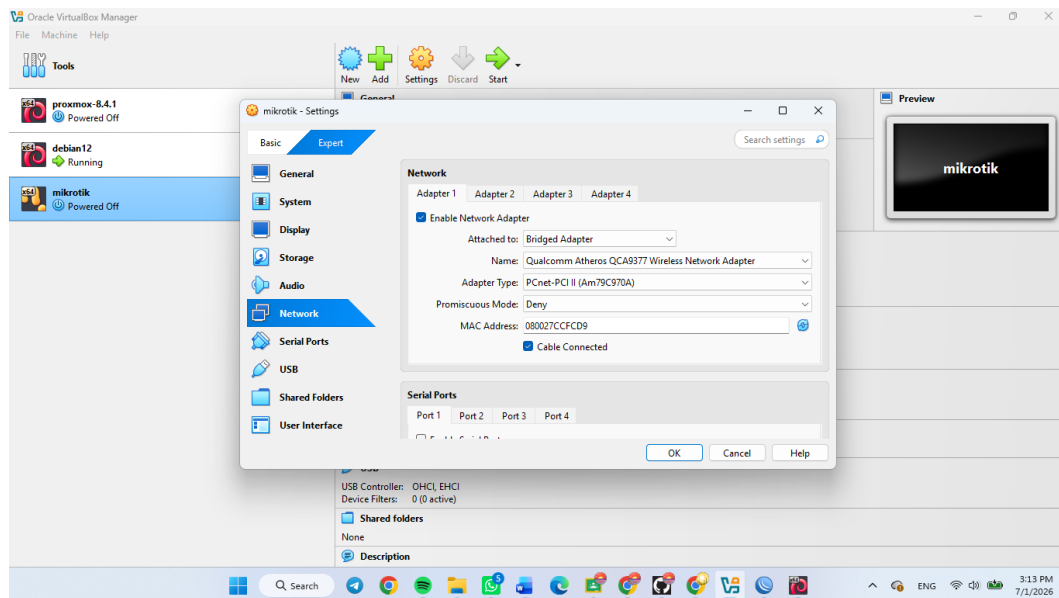
POLITEKNIK PURBAYA

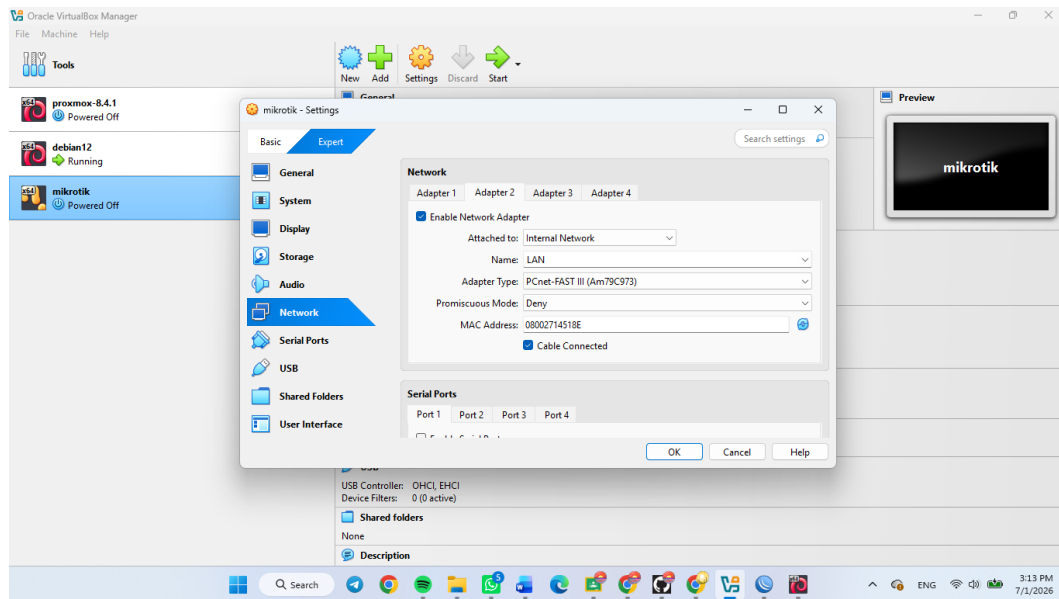
2026/2027

Setelah VM MikroTik selesai dibuat dan file ISO sudah terpasang di **Settings** → **Storage**.

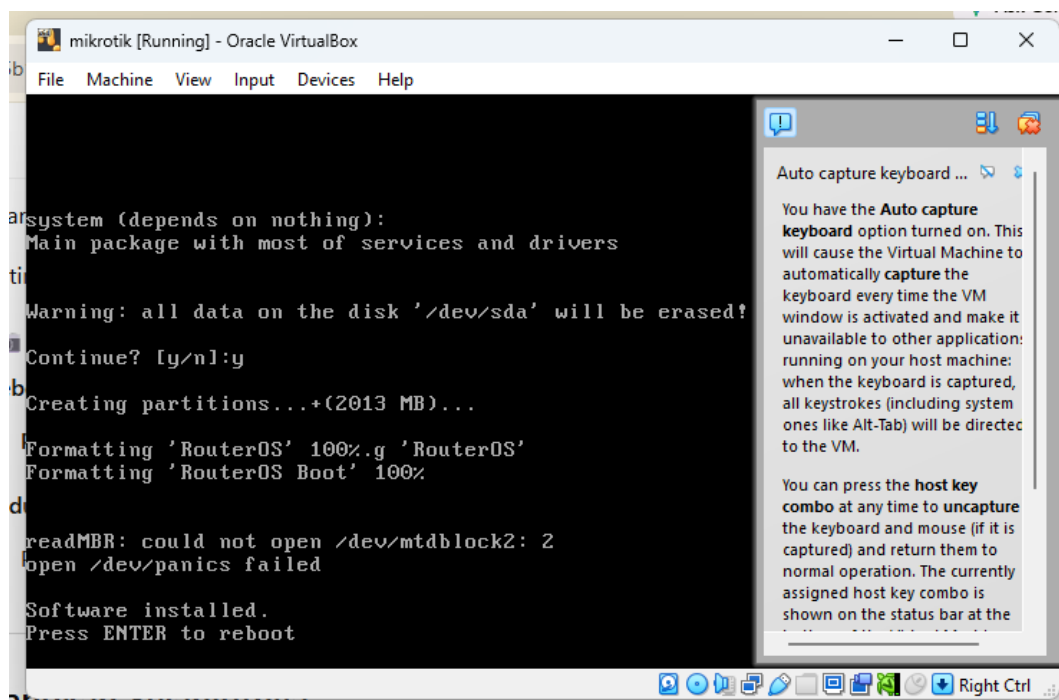


Konfigurasi Network Adapter

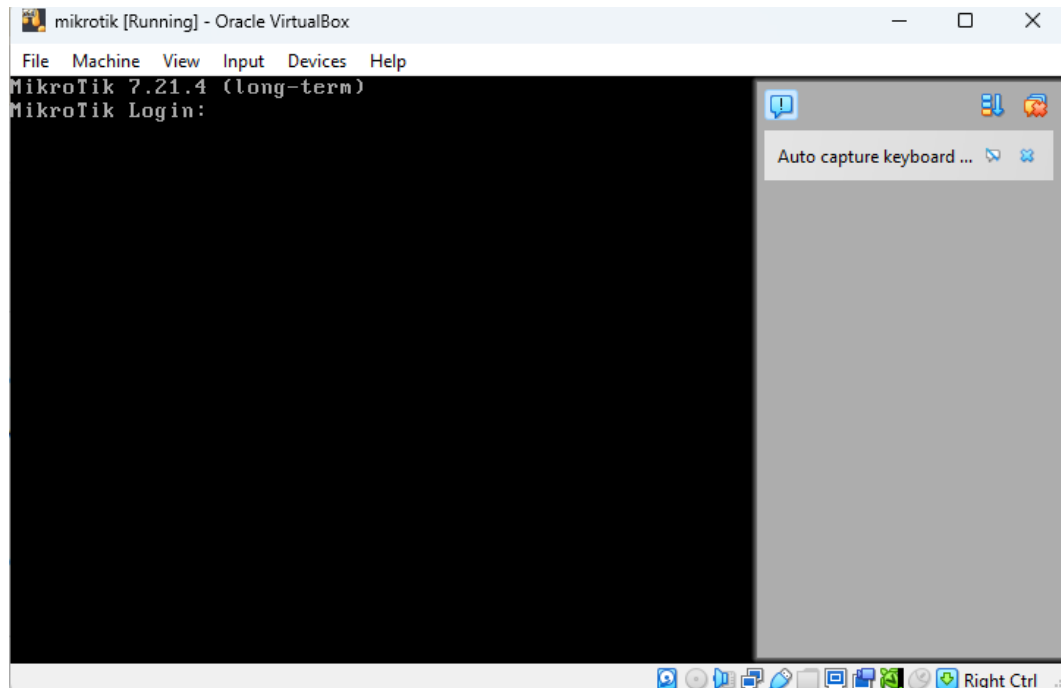




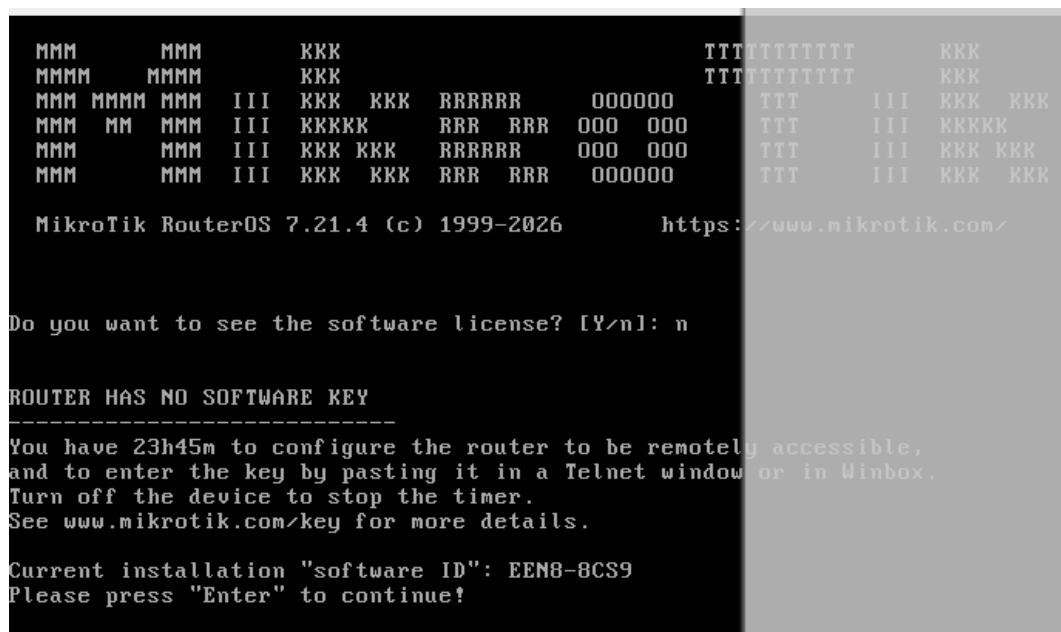
Proses Instalasi RouterOS Selesai



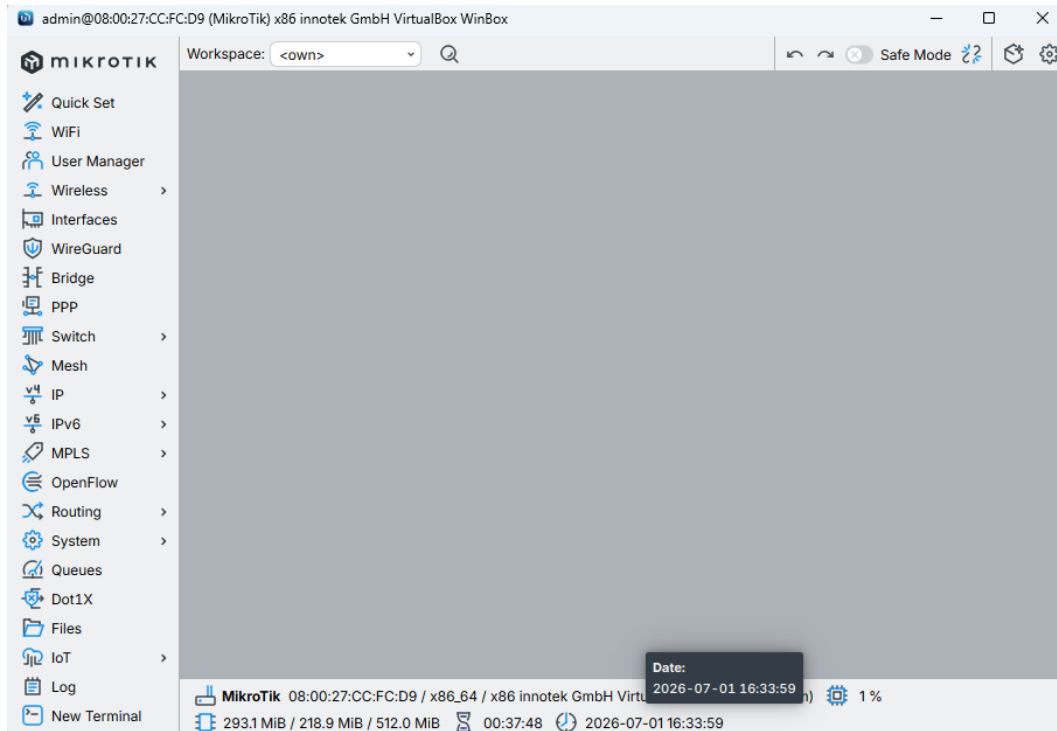
Tampilan Login MikroTik RouterOS



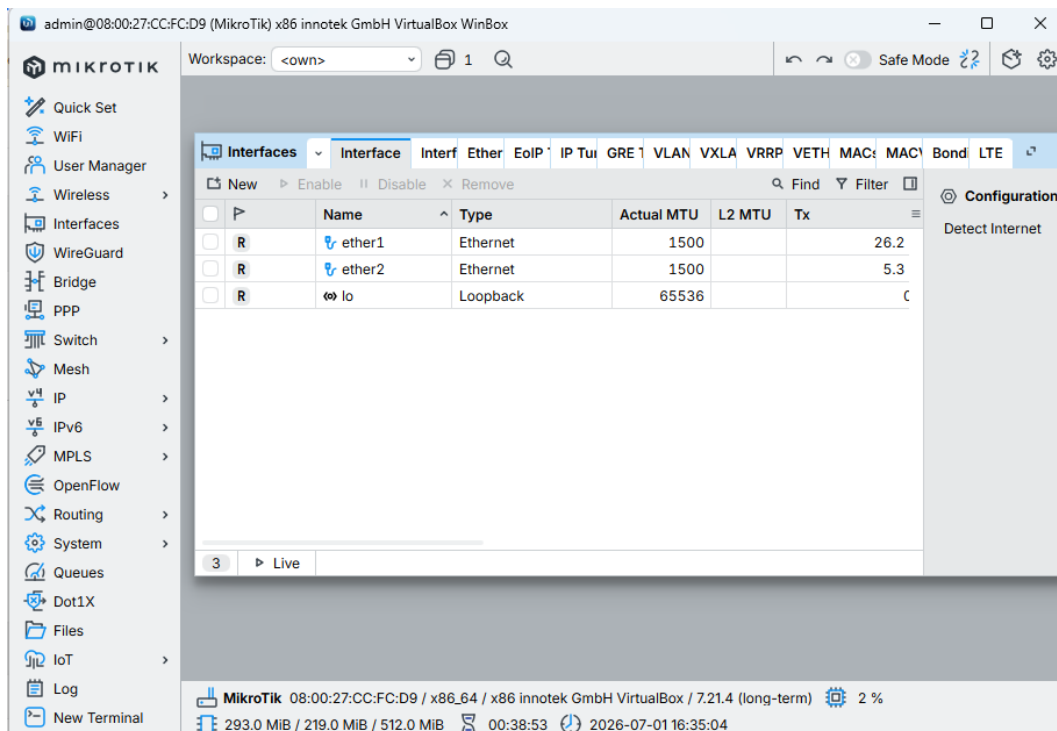
Berhasil Login ke MikroTik RouterOS



Tampilan Awal Winbox Setelah Berhasil Login



Daftar Interface MikroTik



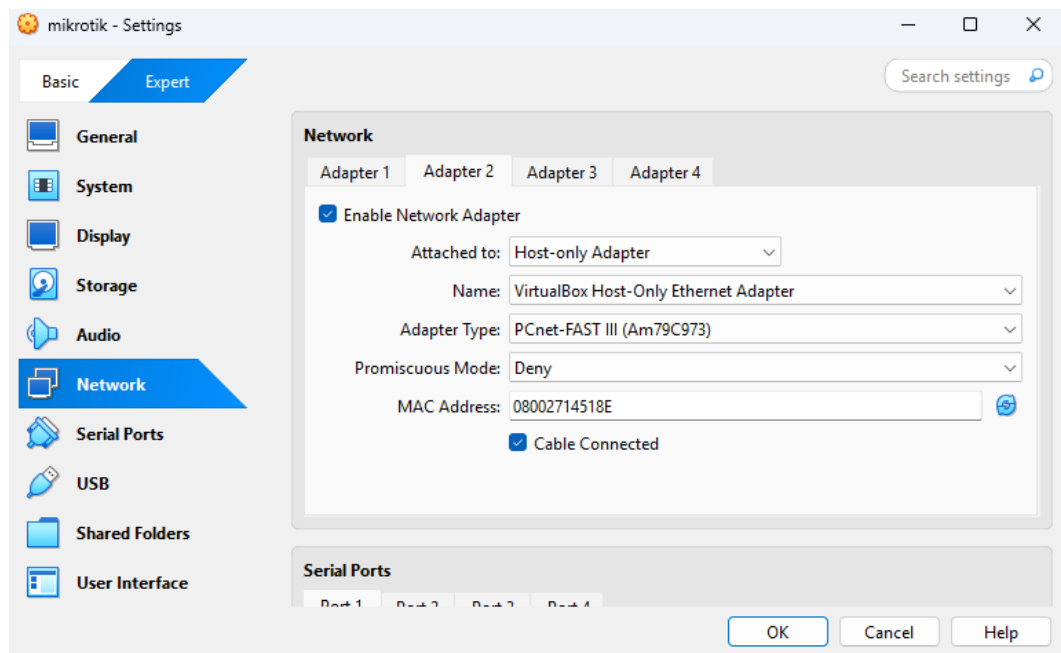
Konfigurasi DHCP Client Berhasil (Status Bound)

```
mikrotik [Running] - Oracle VirtualBox
File Machine View Input Devices Help

4:8D:A9:AA:79 via winbox
2026-07-01 16:45:19 system,error,critical login failure for user admin from 10.0.2.15
4:8D:A9:AA:79 via winbox
2026-07-01 16:45:30 system,error,critical login failure for user admin from 10.0.2.15
4:8D:A9:AA:79 via winbox
2026-07-01 16:51:59 system,error,critical router was rebooted without proper shutdown

[admin@MikroTik] > /ip address print
Flags: D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE, VRF
# ADDRESS NETWORK INTERFACE VRF
0 D 10.0.2.15/24 10.0.2.0 ether1 main
[admin@MikroTik] > /ip dhcp-client print
Columns: INTERFACE, USE-PEER-DNS, ADD-DEFAULT-ROUTE, STATUS, ADDRESS
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes bound 10.0.2.15/24
[admin@MikroTik] > /ip dhcp-client print detail
Flags: X - disabled, I - invalid, D - dynamic
0 interface=ether1 add-default-route=yes default-route-distance=1
default-route-tables=default check-gateway=none use-peer-dns=yes
use-peer-ntp=yes allow-reconfigure=no use-broadcast=both
dhcp-options=hostname,clientid status=bound address=10.0.2.15/24
gateway=10.0.2.2 dhcp-server=10.0.2.2 primary-dns=10.0.2.3
expires-after=23h54m20s
[admin@MikroTik] >
```

Konfigurasi Adapter 2 menggunakan Host-Only Adapter



Konfigurasi IP Address pada interface ether2

```
mikrotik [Running] - Oracle VirtualBox
File Machine View Input Devices Help
(25 messages not shown)
2026-07-01 16:44:36 system,error,critical login failure for user admin from 00:F
1:8D:A9:AA:79 via winbox
2026-07-01 16:44:47 system,error,critical login failure for user admin from 00:F
1:8D:A9:AA:79 via winbox
2026-07-01 16:44:58 system,error,critical login failure for user admin from 00:F
1:8D:A9:AA:79 via winbox
2026-07-01 16:45:09 system,error,critical login failure for user admin from 00:F
1:8D:A9:AA:79 via winbox
2026-07-01 16:45:19 system,error,critical login failure for user admin from 00:F
1:8D:A9:AA:79 via winbox
2026-07-01 16:45:30 system,error,critical login failure for user admin from 00:F
1:8D:A9:AA:79 via winbox
2026-07-01 16:51:59 system,error,critical router was rebooted without proper shu
tdown
2026-07-01 17:06:54 system,error,critical router was rebooted without proper shu
tdown
[admin@MikroTik] > /ip address add address=192.168.3.1/24 interface=ether2
[admin@MikroTik] > /ip address print
Flags: D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE, VRF
# ADDRESS NETWORK INTERFACE VRF
0 D 10.0.2.15/24 10.0.2.0 ether1 main
1 192.168.3.1/24 192.168.3.0 ether2 main
[admin@MikroTik] >
```

Konfigurasi NAT (Masquerade)

```
mikrotik [Running] - Oracle VirtualBox
File Machine View Input Devices Help
8 winbox 8291 tcp main 20
9 api 8728 tcp main 20
10 api-ssl 8729 tcp none main 20
[admin@MikroTik] > /ip address print
Flags: D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE, VRF
# ADDRESS NETWORK INTERFACE VRF
0 D 10.0.2.15/24 10.0.2.0 ether1 main
1 192.168.3.1/24 192.168.3.0 ether2 main
[admin@MikroTik] > /ip address remove [find address="192.168.3.1/24"]
[admin@MikroTik] > /ip address add address=192.168.202.2/24 interface=ether2
[admin@MikroTik] > /ip address print
Flags: D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE, VRF
# ADDRESS NETWORK INTERFACE VRF
0 D 10.0.2.15/24 10.0.2.0 ether1 main
1 192.168.202.2/24 192.168.202.0 ether2 main
[admin@MikroTik] >
[admin@MikroTik] > /ip dns set servers=8.8.8.8,1.1.1.1 allow-remote-request=yes
[admin@MikroTik] > /ip firewall nat add chain=srcnat out-interface=ether1 action
=masquerade
[admin@MikroTik] > /ip firewall nat print
Flags: X - disabled, I - invalid; D - dynamic
0 chain=srcnat action=masquerade out-interface=ether1
line 2 of 2>
```

```
Terminal
[admin@MikroTik] > /ip dhcp-server network print
Columns: ADDRESS, GATEWAY, DNS-SERVER
# ADDRESS          GATEWAY          DNS-SERVER
0 192.168.202.0/24  192.168.202.2    8.8.8.8
                        1.1.1.1

[admin@MikroTik] > /interface print
Flags: R - RUNNING
Columns: NAME, TYPE, ACTUAL-MTU, MAC-ADDRESS
# NAME  TYPE  ACTUAL-MTU  MAC-ADDRESS
0 R ether1 ether      1500  08:00:27:CC:FC:D9
1 R ether2 ether      1500  08:00:27:14:51:8E
2 R lo   loopback  65536  00:00:00:00:00:00

[admin@MikroTik] > /ip dhcp-server network print
Columns: ADDRESS, GATEWAY, DNS-SERVER
# ADDRESS          GATEWAY          DNS-SERVER
0 192.168.202.0/24  192.168.202.2    8.8.8.8
                        1.1.1.1

[admin@MikroTik] > /ip firewall filter add chain=forward protocol=tcp dst-port
=80 content="example.com" action=drop comment="Blok Example HTTP"
[admin@MikroTik] > /ip dns static add name=facebook.com address=0.0.0.0
failure: bad A data: IPv4 address expected
[admin@MikroTik] > /ip dns static add name=www.facebook.com address=0.0.0.0
failure: bad A data: IPv4 address expected
[admin@MikroTik] >
```

```
Terminal
failure: bad A data: IPv4 address expected
[admin@MikroTik] > /ip dns static add name=www.facebook.com address=0.0.0.0
failure: bad A data: IPv4 address expected
[admin@MikroTik] > /ip firewall filter add chain=forward protocol=tcp tls-host
=*.youtube.com action=drop comment="Blok YouTube"
[admin@MikroTik] > /ip firewall filter add chain=forward protocol=tcp tls-host
=*.facebook.com action=drop comment="Blok Facebook"
[admin@MikroTik] > /ip firewall filter add chain=forward protocol=tcp tls-host
=*.instagram.com action=drop comment="Blok Instagram"
[admin@MikroTik] > /ip firewall filter print
Flags: X - disabled, I - invalid; D - dynamic
0      ;;; Blok Example HTTP
      chain=forward action=drop protocol=tcp dst-port=80
      content=example.com

1      ;;; Blok YouTube
      chain=forward action=drop protocol=tcp tls-host=*.youtube.com

2      ;;; Blok Facebook
      chain=forward action=drop protocol=tcp tls-host=*.facebook.com

3      ;;; Blok Instagram
      chain=forward action=drop protocol=tcp tls-host=*.instagram.com
[admin@MikroTik] >
```